

$$E[X_1 + \dots + X_N] = \sum_{n=1}^{\infty} P(N=n) E[X_1 + \dots + X_n | N=n]$$

$$E[X_1 + \dots + X_n | N=n] = E[X_1 + \dots + X_n]$$

$$= \sum_{i=1}^n E[X_i | N=n]$$

$$\{N=n\} = \{X_1 \leq a, X_2 \leq a, \dots, X_{n-1} \leq a, X_n > a\}$$

$$E[X_i | X_1 \leq a, \dots, X_{n-1} \leq a, X_n > a]$$

$$= \begin{cases} E[X_i | X_i \leq a] & \text{for } 1 \leq i \leq n-1 \\ E[X_n | X_n > a] & i=n \end{cases}$$

$$= \begin{cases} E[X_1 | X_1 \leq a] & \text{for } 1 \leq i \leq n-1 \\ E[X_1 | X_1 > a] & i=n \end{cases}$$

$$\Rightarrow E[X_1 + \dots + X_n | N=n] = (n-1) E[X_1 | X_1 \leq a] + E[X_1 | X_1 > a]$$

$$E[X_1 + \dots + X_N] = (E[N] - 1) E[X_1 | X_1 \leq a] + E[X_1 | X_1 > a]$$

$$\text{Let } p = P(X_1 \leq a)$$

$$P(N=n) = p^{n-1} (1-p)$$

$$E[N] = \frac{1}{1-p} \quad (N \text{ is Geometric in } (1-p))$$

$$E[X] = E[X_1 | X_1 \leq a] P(X_1 \leq a) + P(X_1 > a) E[X_1 | X_1 > a]$$

$$= E[x_1 | x_1 \leq a] \phi + (1-\phi) E[x_1 | x_1 > a]$$

$$E[x_1 + \dots + x_N] = (E[N] - 1) E[x_1 | x_1 \leq a] + E[x_1 | x_1 > a]$$

$$= \left(\frac{1}{(1-\phi)} - 1 \right) E[x_1 | x_1 \leq a] + (1-\phi) E[x_1 | x_1 > a]$$

$$= \frac{\left\{ \phi E[x_1 | x_1 \leq a] + (1-\phi) E[x_1 | x_1 > a] \right\}}{(1-\phi)}$$

$$= \frac{E[x_1]}{(1-\phi)} = E[x_1] E[N]$$